Dylan Cunliffe

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Technical Skills

Languages: C, C++, Python, Assembly, SystemVerilog, Verilog, MATLAB
Embedded Systems & Hardware: PCB Design, Analog & Mixed-Signal Design, FPGA Design, STM32, ESP32, CAN/SPI/I2C/UART, RF & Wireless, Signal Integrity, Power Electronics, Piezoelectrics, Energy Harvesting
Software & Tools: Altium Designer, LTSpice, Linux, Quartus, ModelSim, Git, NumPy/scikit-learn, Digital Signal Processing, Machine Learning, Oscilloscopes, Soldering

Education

University of British Columbia — **Expected Graduation: Apr. 2029**
Bachelor of Applied Science in Electrical Engineering — ***CGPA:** 3.95/4.33*

Experience

R&D Engineer, RF & Embedded Systems — May 2026 – Present
e-Radio Inc. — *Vancouver, BC*

- Developed an **ambient-RF location authentication system** verifying a device's physical location from **101-channel FM broadcast fingerprints** (RSSI, multipath), reaching **sub-2% equal error rate** across an 11-site field corpus with **no GPS, internet, or return channel**
- Designed a **drift-suppression pipeline** separating stable location structure from atmospheric propagation drift, **cutting equal error rate 42%** (26.8% → 15.5%) across a seven-site validation harness
- Deployed the **int8-quantized model to an STM32N6 NPU** and integrated the verifier into a live over-the-air demo, achieving **sub-millisecond trust decisions** and **39/39 correct accept/refuse outcomes** against tampered, replayed, and off-site attacks
- Architected a **low-cost data-collection fleet**: **Si4732 receiver PCB** in Altium, **ESP32 captive-portal provisioning**, offline SD logging, and cloud ingest schema for autonomous multi-site RF collection

Bicycle Mechanic — May 2025 – Sep. 2025
Spokes Bicycle Rentals — *Vancouver, BC*

- **Diagnosed and repaired mechanical failures** on 1000+ bicycles, completing 1700+ maintenance entries over 4 months by optimizing workflow for high-volume constraints.

Technical Projects

Through-Wall Acoustic Sensor | *LTSpice, Analog Design, Piezoelectrics, Bench Testing* — Apr. 2026 - Present

- Engineered a through-wall **acoustic power and data transfer** system to enable battery-less wireless sensors inside sealed steel industrial pressure vessels at 15% power efficiency
- Designed **analog drive circuit, LC matching network, and harvesting receiver (voltage doubler, boost converter)** to extract power from acoustic wave transmission through a mild steel plate via PZT-5A transducers
- Conducted **bench characterization** using oscilloscopes and function generators to extract BVD equivalent circuit parameters, identify mechanical resonance, and perform **maximum power point tracking** (MPPT)

Mixed-Signal Doppler Radar System | *STM32, LTSpice, Analog Design, Active Filters* — Feb. – Mar. 2026

- Designed and **simulated** an active analog signal chain in LTspice, utilizing an active bandpass filter and rail-to-rail op-amps to cleanly amplify microvolt-level Doppler shifts from a 10.5 GHz X-band radar module
- Engineered a 4-layer **mixed-signal PCB** in Altium Designer, implementing strict floorplanning to isolate noisy high-speed digital domains (I2C/UART) from the sensitive analog filters
- Developed bare-metal C firmware for an STM32G4 to calculate velocity and log telemetry, **executing physical bring-up** with oscilloscopes to validate LDO stability and analog-digital **signal integrity**

Automotive Sensor Telemetry PCB | *Altium Designer, CAN, Power Electronics* — Nov. – Dec. 2025

- Engineered a 2-layer PCB in Altium Designer integrating an STM32G0 MCU for multi-protocol telemetry **(CAN, SPI, I2C, UART)**, with **differential pair routing** for high-speed CAN, **thermal relief**, and strict **EMI zoning** to minimize ground loops
- Designed a high-efficiency power stage using a DC-DC Buck Converter with **input filtering** and **reverse polarity protection** to step down 12V vehicle power to a stable 3.3V logic rail

Edge AI Traffic Light Control System | *Python, YOLOv8, Nvidia Jetson, Embedded AI* — Oct. – Dec. 2025

- Engineered an **embedded vision system** using the NVIDIA Jetson Orin Nano to detect vehicles with YOLOv8n and dynamically control a **hardware traffic light system** in real time using an FSM traffic controller
- Achieved stable **real-time inference** (∼30 FPS) using CUDA-accelerated preprocessing and TensorRT model optimization for efficient deployment on embedded Linux

View more projects on my portfolio website...